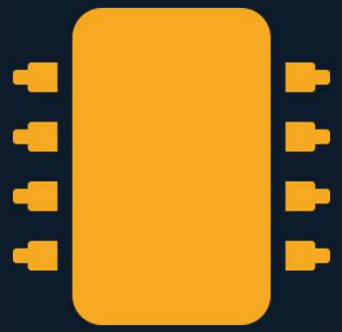
NoEtherX Labs

XYTRIX

Gate-Level Simulation (GLS)

Analysis & Timing Verification

Zero Delay vs SDF Delay Simulation



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Project Overview

Xytrix — GLS Objectives & Scope

Objective of GLS

- Functional verification after synthesis
- Timing-aware verification using SDF
- Comparison of Zero Delay GLS vs SDF GLS
- Log analysis using Cadence Xcelium
- Post-synthesis netlist correctness

Verification Stages

Verification Stage	Purpose
RTL Simulation	Functional validation
GLS	Netlist verification
SDF GLS	Timing-aware verification

Zero Delay

Fast functional check

SDF GLS

Silicon-accurate timing

Xcelium

Cadence simulation engine

Why GLS is Important

Post-Synthesis Verification Necessity



Verification flow — GLS bridges synthesis and physical signoff

- Detects post-synthesis issues
- Reset & clock tree behavior
- Validates real silicon behavior
- DFT logic verification
- Timing correctness verification
- Glitch detection

Reference: Cadence GLS Flow Guide

Types of GLS

Simulation Delay Models

Zero Delay

- No timing annotation
- Fastest simulation
- Specify blocks ignored
- Early verification

Timing: None
Speed: Fast
Usage: Functional checks

Unit Delay

- Uniform unit delays
- Detects race conditions
- Better than zero-delay
- Not silicon-accurate

Timing: Approx.
Speed: Medium
Usage: Race detection

SDF Delay

- Back-annotated delays
- Silicon-correlated
- Setup/hold checking
- Production signoff

Timing: Accurate
Speed: Slow
Usage: Signoff

Type	Timing	Speed	Usage
Zero Delay	No	Fast	Functional checks
Unit Delay	Approximate	Medium	Race detection
SDF GLS	Accurate	Slow	Signoff verification

GLS Verification Flow

End-to-End Simulation Pipeline

1. **Netlist Generation**
   Post-synthesis gate-level netlist
2. **SDF Generation**
   Timing annotation from place & route
3. **GLS Execution**
   xrun simulation with library models
4. **Timing Analysis**
   Setup/hold checks, SDF annotation
5. **Debugging**
   Warning analysis, loop detection
6. **Signoff**
   Clean simulation, tapeout ready

Required Inputs

- Gate-level netlist (.v)
- Liberty library models (.lib / .v)
- SDF timing file (.sdf)
- Testbench with assertions
- SDF command file (optional)

Reference: Cadence GLS Flow Guide

Experimental Setup

Tools, Files & Environment Configuration

Tools Used

- Cadence Xcelium Simulator
- Verilog Gate-Level Netlist
- SDF Timing Annotation File
- Cadence Standard Cell Library

Cadence Xcelium

Industry-standard simulation engine

Files Used

File	Purpose
counter_netlist.v	Gate-level netlist
delays.sdf	Timing annotation
counter_test.v	Testbench
slow_vdd1v0_basicCells.v	Library models

Parameter	Value
Simulator	Cadence Xcelium
Timescale	1ns / 10ps
Netlist	Gate-level Verilog
Library	slow_vdd1v0_basicCells

24	16	12	144
Modules	UDPs	Registers	Timing Checks

Zero Delay GLS

Functional Verification Without Timing Annotation

Characteristics

- No timing annotation required
- Specify delays are ignored
- Faster runtime execution
- Functional-only verification
- No setup / hold checking

xrun Command — Zero Delay GLS

```
xrun -clean counter_netlist.v counter_test.v \
  -mess -access +rwc \
  -v slow_vdd1v0_basicCells.v \
  -timescale 1ns/10ps -nospecify
```

~3×	0	Fun.
Faster than SDF GLS	Timing violations	Verification only



-nospecify — disables all timing checks and specify blocks

Zero Delay GLS — Log Analysis

Xcelium Simulation Log Review

⚠ **Xcelium Log Warning – xmelab**

```
xmelab: *W,SDFSKPA:
    Ignore $sdf_annotate because of -nospecify option.
```

Analysis Points

- SDF file is skipped — annotation ignored
- Timing checks are disabled via -nospecify
- Only functional verification is performed
- Simulation completes without timing violations

Parameter	Value
Modules	24
UDPs	16
Registers	12
Timing Checks	144 (disabled)
Timescale	10ps

SDF GLS

Standard Delay Format — Timing-Accurate Simulation

Characteristics

- Real timing delays applied
- Timing checks enabled
- Silicon-accurate verification
- Slower simulation runtime
- Detects setup / hold violations
- Preferred for tapeout signoff

xrun Command — SDF GLS

```
xrun -v slow_vdd1v0_basicCells.v \
  counter_netlist.v counter_test.v \
  -mess -access +rwc \
  -timescale 1ns/10ps \
  -sdf_cmd_file sdf_cmd_file
```

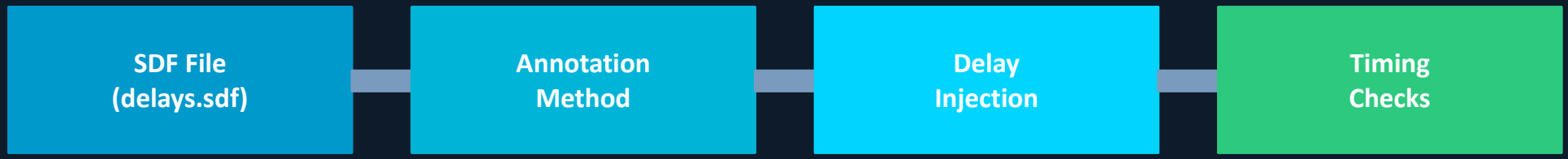
100%	144	67
Path delay coverage	Timing checks	Interconnects annotated



Key Concept: Back-annotation of delays from place & route into simulation

SDF Annotation Process

Back-Annotation Flow & Methods



$sdf_annotate

- Inline Verilog system task
- Called inside testbench
- Targets specific module instances
- Fine-grained annotation control

```
$sdf_annotate("delays.sdf", dut);
```

-sdf_cmd_file

- Command-line driven approach
- External SDF command file
- Multiple instances supported
- Preferred for automation

```
-sdf_cmd_file sdf_cmd_file
```

SDF GLS — Log Analysis

Xcelium Annotation Summary

✓ **Xcelium Log — SDF Annotation**

```
Reading SDF file from location "delays.sdf"
SDF annotation completed successfully.
xmsim: *I,SDFANN: Annotation statistics: see report.
```

Annotation Statistics

Parameter	Status
Path Delays	100% Annotated
$setuphold	100% Annotated
$recovery	100% Annotated
$removal	100% Annotated

Simulation Metrics

Parameter	Value
Path Delays Annotated	57
Timing Checks	144
Interconnects	67
Annotation Coverage	100%



Full SDF annotation achieved — all timing paths and checks successfully back-annotated

Delay Models in GLS

Transport vs Inertial Delay Behaviour

Transport Delay

- Propagates ALL pulses
- No glitch filtering
- Any pulse width propagated
- Used for interconnects
- Wires and buffers

Inertial Delay

- Removes narrow glitches
- Pulse width threshold applied
- More realistic gate behavior
- Used for logic gates
- Default in GLS

Feature	Transport	Inertial
Glitch Filtering	No	Yes
Pulse Rejection	No	Yes
Usage	Interconnects	Gates

Timing Checks in GLS

Setup, Hold & Sequential Constraints

Setup	Hold	Recovery	Removal	Width
Data must be stable before clock edge by tSU	Data must remain stable after clock edge by tH	Async reset must deassert before clock edge	Async reset must hold after clock edge	Minimum pulse width for clock and data signals

Timing Violation — xmsim Log Example

```
Warning! Timing violation — $setuphold<setup>  Time: 50 ns
         Violation: 0.3ns  Required: 0.8ns  Current: 0.5ns
```

Violation Analysis

- Indicates timing path failure in silicon
- Setup violation → hold/route fix required
- Hold violation → buffer insertion needed
- Requires STA correlation and ECO

144

Total Timing Checks in Xytrix Design

(disabled in Zero Delay — active in SDF GLS)

Common GLS Issues

Known Warnings, Errors & Debugging Guidance

Zero-Delay Loop

- Missing delay on feedback path
- Causes infinite oscillation
- Use -gateloopwarn to detect

SDFNEP Warning

- SDF path doesn't match netlist
- Annotation failure on cell
- Check instance path names

Timing Violation

- Setup / hold failure
- Indicates timing path issue
- Requires STA correlation

Annotation Mismatch

- SDF version mismatch
- Cell name differences
- Verify library compatibility

Issue	Cause	Effect
SDFNEP	Path mismatch	Annotation failure
Zero-delay loop	Missing delay	Infinite oscillation
Timing violation	Setup/Hold failure	Incorrect timing
Glitch error	Inertial mismatch	False violations

Important GLS Switches

xrun Simulator Options & Debugging Flags

`-nospecify`	Disables timing checks and specify blocks *Usage: Zero Delay GLS*
`-sdf_cmd_file`	Loads SDF command file for annotation *Usage: SDF GLS*
`-sdf_verbose`	Generates detailed SDF annotation log *Usage: Debugging*
`-gateloopwarn`	Warns on zero-delay feedback loops *Usage: Loop detection*
`-pulse_r`	Pulse rejection threshold control *Usage: Inertial delay*
`-notimingchecks`	Disables all timing checks globally *Usage: Fast functional sim*



Practical debugging tip: combine -sdf_verbose + -gateloopwarn for comprehensive annotation and loop diagnostics

Zero Delay vs SDF GLS

Comprehensive Feature Comparison

Feature	Zero Delay GLS	SDF GLS
Timing Annotation	No	Yes
Functional Verification	Yes	Yes
Timing Verification	No	Yes
Simulation Speed	Faster	Slower
Glitch Detection	Limited	Accurate
Silicon Correlation	Low	High
Setup / Hold Checks	Disabled	Enabled
Usage	Early verification	Signoff

Results & Observations

Xytrix GLS Simulation Findings

Zero Delay GLS

- Faster simulation execution (~3x)
- Functional-only analysis confirmed
- SDF annotation skipped (SDFSKPA warning)
- No timing violations reported
- Suitable for early netlist validation
- Cannot detect timing failures

SDF GLS

- 100% path delay annotation achieved
- All 144 timing checks active
- Setup/hold violations detectable
- Better silicon correlation
- Accurate glitch detection
- Required for tapeout signoff



Key Learning: Timing-aware SDF GLS is critical before tapeout — functional verification alone is insufficient

Final Conclusion

Achievements

- Understood GLS verification workflow
- Performed Zero Delay GLS simulation
- Performed SDF GLS with annotation
- Analyzed timing annotation statistics
- Compared transport vs inertial delay models
- Studied debugging switches and techniques

Final Takeaway

GLS remains an essential step in VLSI verification despite advances in STA and formal verification tools.

It bridges the gap between RTL intent and physical silicon behavior — catching real-world issues that static analysis alone cannot.

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